

BITS PILANI, PILANI CAMPUS
ACADEMIC-UNDERGRADUATE STUDIES DIVISION
First Semester 2025–2026
Course Handout

Date: 03.08.2026

Course No. : BITS U107
Course Title : Introduction to Engineering
Course Instructor : Prof. Ravindra Dwivedi / Prof. Somak Chatterjee
Instructor-in-charge : Prof. Ravindra Dwivedi

Scope and Objective of the Course:

Introduction to the different engineering disciplines (e.g., civil, mechanical, electrical, chemical, computer, etc.), Historical development of engineering, Role of engineers in society, Ethical considerations in engineering, Problem-solving methodologies, Basics of physics and mechanics, Introduction to engineering design and the design process, Basic principles of thermodynamics, Sustainable engineering practices, Environmental impact assessment, Engineering and its influence on society, Business and Engineering.

Course Pre- /Co- requisite (if any) & Catalogue / Bulletin Description:

Textbook (TB)

TB: Saeed Moaveni, "Engineering Fundamentals: An Introduction to Engineering", Cengage Learning, 5th Edition, 2015.

Reference Books (RBs)

R1: N/A

Course Plan / Schedule

Week No	Learning objectives	Contents	Chapter No (TB)
#1 (2hours)	Introduction to the different engineering disciplines	Disciplines (civil, mechanical, electrical, chemical, computer, architecture, etc.) Sub-disciplines Specializations	Notes
#1 (2hours)	Historical development of engineering	Engineering in the B.C. Engineering in the A.D. first millennium Engineering Today	Notes
#2 (2hours)	Role of engineers in society	Education, Practice, Certification Influence in Government Influence in Society	1,2
#2 (2hours)	Ethical considerations in engineering	Role and types of engineering communication Engineering solution methodology Engineering Ethics	4, 5
#3 (4hours)	Introduction to engineering design and the design process	Design process Design Considerations Teamwork Project Management Engineering Standards and Codes	3
#4,5 (6hours)	Basics of physics and mechanics	Dimensions and Units Newton's Laws	6-9

#5,6 (6hours)	Basic principles of thermodynamics	Units Calorific value Comfort Energy management	11, 13
#7,8,9 (12hours)	Problem-solving methodologies	Understanding needs, requirements Hypothesis, testing and prototypes Demonstration, Life-cycle Product, Project and Service Approaches Creativity	Notes
#10 (2hours)	Sustainable engineering practices	Management of inputs Management of processes Management of outputs Use of Minerals and Materials	13, 17
#10 (2hours)	Environmental impact assessment	Impact to air, soil, water Impact to Life and Health (HSE practice)	Notes
#11 (2hours)	Engineering and its influence on society	Adapting to and use of Environment Changing Landscapes	Notes
#11 (2hours)	Business and Engineering	Engineering economics Business models Engineering as a backbone to business	Notes

Course Learning Objectives:

CLO1 Awareness, understanding and identifying engineering disciplines, sub-disciplines

CLO2 Understanding the fundamentals of managing engineering processes and teams

CLO3 Making, questioning and evaluating engineering calculations

CLO4 Understanding the pivotal role of engineering in business

CLO5 Awareness of trends in engineering, challenges to society

Evaluation scheme:

EC No.	Evaluation Components	Nature of Component	Duration	Weightage	Date & Time	Venue
1	Midterm Exam	Open-Book*	60 minutes	20 %		To be announced later
2	Assignments**	Open-Book/Take-Home/Tutorial		60 %	TBA	
3	Comprehensive Exam	Closed-Book	90 minutes	20 %		

* Only prescribed textbook(s) and handwritten notes are permitted

**Assignments: During the semester, each student will work individually or in a team on the given assignment. These assignments will be based on the broad topics covered in the course and will be announced in the class from time to time. Each individual/team of students will work on these assignments. The deadline for submitting these assignments will also be announced in the class itself.

Mid-semester Grading:

Mid-semester grading will be displayed after about 35% of evaluation components are completed. (Refer to the Academic Calendar for schedule).

Note: A student is likely to get “NC”, if he / she doesn’t appear / appear for the sake of appearing for the evaluation components / scoring zero in pre-comprehensive exam total.

Makeup and Attendance policies:

Make-ups: are not given as a routine. It is solely dependent upon the genuineness of the circumstances under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge (I/C). The decision of the I/C in the above matter will be final.

Attendance: Every student is expected to be responsible for regularity of his/her attendance in classrooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. A student should have **a minimum of 60% of attendance** in the course to be eligible to appear for the Comprehensive Examination in the course. For the students under the purview of Academic Counseling Board (ACB), the Board shall prescribe the minimum attendance requirement on a case-to-case basis. Attendance in the course will be a deciding factor in judging the seriousness of a student which may be directly / indirectly related to grading.

CCH Hours: TO BE DETERMINED BY ENTIRE SCHEDULE OF SECTION

General instructions:

Students should come prepared for classes and carry the textbook(s) or material(s) as prescribed by the Course Faculty to the class.

Notices:

All notices will be displayed on Nalanda.

Academic honesty and academic integrity Policy:

Academic honesty and academic integrity are to be maintained by all students throughout the Semester, and no type of academic dishonesty is acceptable.

**Instructor-In-charge
BITS U107**

Instructors Contact Details:

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